

Benjamin Wrinn

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EDUCATION

University of Pennsylvania - The School of Engineering and Applied Science

Philadelphia, PA

Candidate for BSE in Mechanical Engineering (3.25/4.0 UW)

May 2029

Oregon Episcopal School - High School

Portland, OR

- *Cumulative GPA: 3.9/4.0 (UW); SAT - 1570 | Math 780 | Writing 790*

May 2025

RELEVANT EXPERIENCE

Vero Electric / PowerTown

Philadelphia, PA

Lead Mechanical Engineer

June 2026 – Present

- Created a custom, **240kWh/120kW BESS enclosure**: from Solidworks to physical assembly. Designed to withstand -20 - 55°C ambient extremes and pass *UL safety standards*. (Certification Pending)
- Optimized for low-cost manufacturing; validated architecture through Solidworks **Thermal** and **Static** simulations.
- Routed a 64-cell-series BMS node PCB in KiCad compatible with Hithium battery enclosures, cutting node integration time by enabling drop-in fit with existing pack architecture

Penn Electric Racing Formula SAE EV Team

Philadelphia, PA

Mechanical Team

Sept 2025 – Present

- Designing **two-motor electric dynamometer** in preparation for 4WD and Design Competition
- Engineering a servo-actuated Drag Reduction System projected to cut straight-line drag by **37.7%**, improving lap times by **2 seconds**.
- Modeled and detailed chassis/suspension components in SolidWorks per GD&T standards, then manufactured carbon fiber parts via resin infusion and wet layup at sponsor facilities. Produced KeyShot renders used in sponsor and design-review presentations

American Rocketry Challenge

Portland, OR

Team Host, Hardware and Media Lead

Sept 2022 – August 2025

- Built a dynamic air brake system that autonomously adjusted drag using a Python lookup table, landing the rocket **within 7ft of target apogee**: placing **1st** in State, **20th** nationally and **Top 5** in Presentation
- Designed the control PCB and validated drag predictions using CFD (Solidworks) before flight testing
- Grew team Instagram by +5.5k followers and +350k interactions as Media Lead, driving sponsor visibility and recruitment

Children's Cancer Therapy Development Institute

Portland, OR

Student Research Assistant

June 2023 – June 2025

- Wet-lab experience in both CAM and Bioluminescence assays; Co-authored research manuscript awaiting publication on CAM Assay drug trials.
- 3D modeled, designed and built a prototype for machine vision on quail embryo drug trials, using IR motion detection to quantify embryo motion and time of death; cc-TDI has continued to develop this towards a more economical drug screening product to pair alongside mouse clinical trials.
- Filmed and edited quail instrument instruction videos for customers.

ADDITIONAL

- **Skills:** CAD (SolidWorks – parts, assemblies, drawings, GD&T), Manufacturing (CNC Machining, 3D Printing, Laser Cutting, Wet Layups), Simulations (ANSYS, Star CCM+, SolidWorks FEA, Luminary Cloud), Kicad/Altium PCB Design, Python, Blender, Davinci Resolve, Chinese (Mandarin)
- **Courses:** Linear Algebra, Calc 1-3, Physics 1+2, Chemistry 1, Python, Mechanical Design, Intro to Bioengineering
- **Activities:** USA National Kung Fu Team (Gold, Silver Medalist @ 2023 IWUF, Youth Coach), William Temple House Volunteer (stocked 3700+ lb of pantry items, fulfilled 120+ orders, coded menu translator in Python)